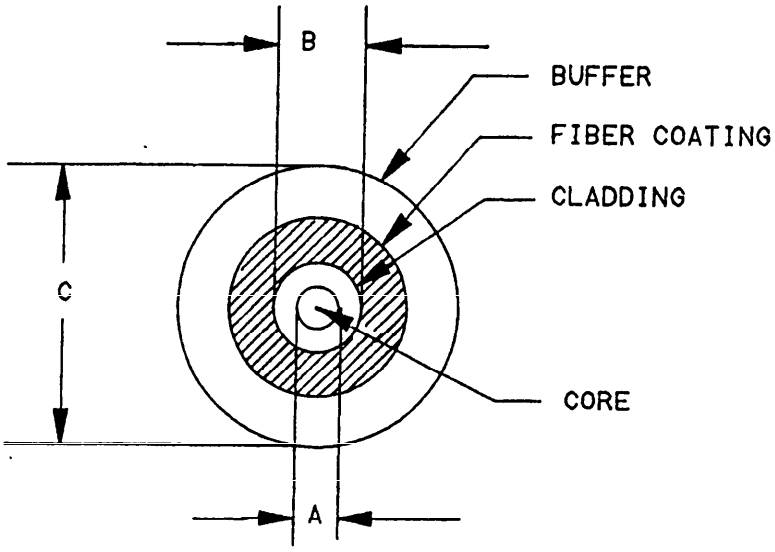
MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTICS, TYPE I, CLASS 2, COMPOSITION A,
ATTENUATION RANGE A, SINGLE FIBER FOR

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables (fibers) described herein shall consist of this specification and the latest issue of DOD-C-85045.



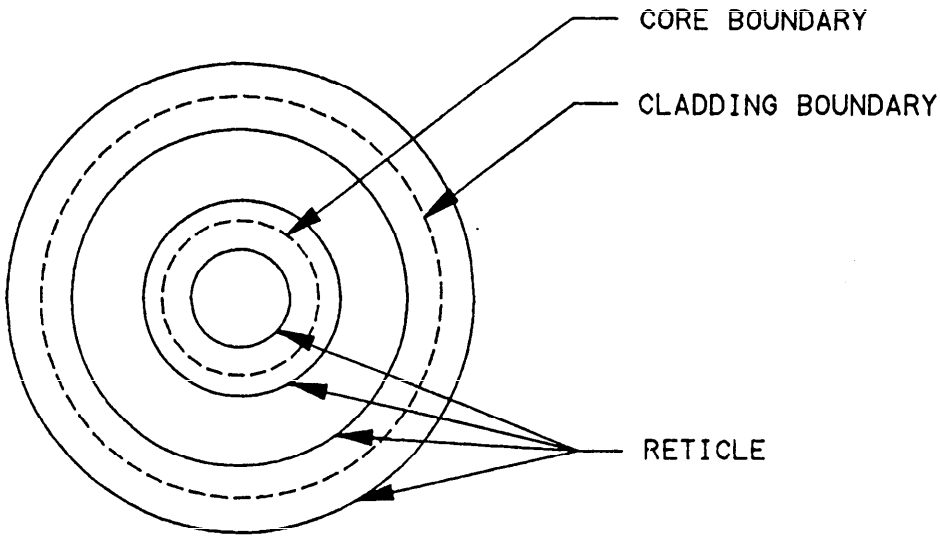
PART NUMBER	DIMENSION (μm)		
	A	B	C
D85045/1-1	50 ±4	125 ±4	900 ±50
D85045/1-2	50 ±4	125 ±4	500 ±50

NOTE: Dimensions are in micrometers.

FIGURE 1. Dimensions and configuration of tight buffered optical fiber construction.

FSC 6015

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CIRCLE	DIAMETER OF CIRCLE
INNER	46 μ m
SECOND	54 μ m
THIRD	121 μ m
FOURTH	129 μ m

NOTE: Dimensions are in micrometers.

FIGURE 2. Tolerance fields.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Core and cladding material: Composition A.

Fiber coating: Silicone elastomer or functional equivalent.

Buffer material: Thermoplastic polyester elastomer or functional equivalent.

Buffer and coating strippability: The buffer and coating material shall be easily and cleanly removable with mechanical strippers without damaging the optical fibers. No surface scratches or defects shall be visible under 10X magnification. Any residual buffer or coating material shall be easily wiped off without the use of chemcials.

Tensile strength of fiber: 690 MPa continuous proof test.

Attenuation: Test length L_1 = 1.000 km minimum measured to ± 1%; test length L_2 = 2.00 m ± 10 cm; -55° to +85°C; the attenuation shall not exceed 5.5 dB/km (820-890 nm); 2.5 dB/km (1100 - 1300 nm).

Optical fiber bandwidth (fiber transfer function): Test length L_1 = 1.000 km minimum, measured to ± 1%; test length L_2 = 2.00 m ± 10 cm; -55° to + 85°C; the optical fiber bandwidth shall be a minimum of 400 MHz-km (820 - 890 nm), 600 MHz-km (1100 - 1300 nm).

Radiation pattern measurement: 90% power; 1.000 km minimum length; 820 - 890 nm, 1100 - 1300 nm. The exit angle shall be 0.20 ± 0.02 radian.

Core and cladding boundaries: Defined by the tolerance fields of figure 2.

Pulse Spreading: Not applicable:

Flammability: Burn time - 30 seconds maximum; flame travel - 10 cm maximum.

Fluid immersion: After testing, there shall be no change in diameter and there shall be no cracks, splits, or voids in the exterior surface. Fluids, temperature, and immersion time shall be as specified below:

Fluid	Temperature (°C)	Immersion time (hours)
Lubricating Oil MIL-L-23699	48 - 50	20
Hydralic Fluid MIL-H-5606	48 - 50	20
Isopropyl Alcohol TT-I-735	20 - 25	168
Turbine Fuel MIL-T-5624	20 - 25	168
Automobile Gas MIL-G-3056	20 - 25	168

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Preparing activity:
Air Force - 85

(Project 6015-0001-1)

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